Lab 7: Fuzzing for Input Validation Bugs

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Step 2: Running the buggy processor.py (python processor.py)

```
PS C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301> python processor.py
==== Running sanitize_meter_id ====
Enter a meter ID with prefixes (!,@,#,$,%): !@#abc123
Sanitized Meter ID: ABC123

==== Running parse_consumption_readings ====
Enter consumption readings (e.g. 10,25,30): 10,,30
Traceback (most recent call last):
  File "C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301\processor.py", line 35, in <module>
    main()
    ~~~~^^
  File "C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301\processor.py", line 26, in main
    readings_list = parse_consumption_readings(csv_input)
  File "C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301\processor.py", line 9, in parse_consumption_readings
    return [int(r) for r in readings]
            ~~~^^^
ValueError: invalid literal for int() with base 10: ''
PS C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301>
```

Step 4: Running test_processor.py against buggy code - tests failing

```
    +----------------------------------
============================================== warnings summary ===============================================
..\..\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Tanishk\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================== short test summary info ==========================================
FAILED test_processor.py::test_sanitize_meter_id_no_crash - AssertionError: sanitize_meter_id crashed with: 'NoneType' object has no attribute 'strip'
FAILED test_processor.py::test_parse_consumption_readings_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_format_billing_summary_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
======================================== 3 failed, 1 warning in 5.12s ========================================
PS C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301>
```

Step 5: Running the fixed processor.py (python processor.py)

```
PS C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301> python processor.py
==== Running sanitize_meter_id ====
Enter a meter ID with prefixes (!,@,#,$,%): !@#abc123
Sanitized Meter ID: ABC123

==== Running parse_consumption_readings ====
Enter consumption readings (e.g. 10,25,30): 10,,30
Parsed Readings: [10, 30]

==== Running format_billing_summary ====
Enter account record (Name:Usage:Zone): Tanishk:50:ZoneA
Billing Summary: Customer Tanishk billed 100 USD in ZoneA
PS C:\Users\Tanishk\SE-lab-PES1UG24AM301\lab-7_PES1UG24AM301>
```

Step 6: Re-running test_processor.py against fixed code - tests passing

